

Smart networks for control

Home, factory, and auto equipment that heeded the commands of embedded microprocessors will soon obey distributed networks

Picture your office building early one morning, some years hence. You get in by sliding your badge through an access system's card reader. Instantly, the lights leading to your office flicker on. As you push open the office door, lights and temperature adjust to the settings you picked yesterday. Meanwhile, the process control network in the manufacturing area sends your computer the latest statistics.

Abruptly, a fire alarm system warns you of smoke and fire in a section of the factory floor. It also alerts the local fire department by phone, and your heating and air conditioning die. The process control network shuts down the manufacturing line, the access control system lists who is where in the building for the fire department, and the lighting control network turns on all lights along the routes to the emergency exits. All this happens painlessly and flawlessly because the control networks interoperate, or less technically, interact.

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The vision may be closer to realization than is generally suspected. In the 1980s, control systems based on embedded microprocessors took on a growing number of systems and products: building, home, and factory automation, as well as automobiles and security systems. But because embedded control systems have practical limits, such as low performance-cost ratios, and inherent problems, such as single points of failure, the demand arose for higher functionality and lower cost, prompting the development of distributed control networks.

Many efforts have since been under way worldwide to establish control networking standards for a range of industries. To name just two, BACnet serves building automation, and CAN is for automotive applications. All define networks that take over tasks formerly done by centralized embedded control systems, but each is different.

NETWORKS COMPARED. Data networks and control networks both divide up the previously centralized brain of the system and relocate it into a set of smaller interconnected processors. But the similarities end there [Fig. 1].

Generally speaking, data networks use large data packets and relatively infrequent transmission rates, with high data rates to support the transmission of large data files. Control networks, in contrast, must shuttle countless small but frequent packets among a relatively large set of nodes.

A data network could serve as a control network—much as a highway can be used

as an airport runway. It will work, but it is not the right fit, and the problems appear when you begin to put demands on the system.

For example, the throughput of an Ethernet local-area network (LAN) drops off suddenly at 35 percent of bandwidth utilization. Yet high loads are common when a network is used for control, because network traffic usually depends directly on external events being monitored by the nodes. Often multiple nodes are needed to send simultaneous packets based on one or more external events. This, coupled with the large number of nodes often present, translates into frequent periods of high traffic for the network, which therefore needs high throughput to work properly.

Nonetheless, data networks and control networks can work together. In integrated form, they could fulfill the need for a new breed geared toward total networking. Such hybrids would do more than manage the data flow between PCs and the central file server in an office building—they would also permit users to control and monitor the environment in which each works, all from the same PC with the same user interface. Cross-linking the two types of networks could be the natural extension of networking in the future.

CONTROL NET ARCHITECTURE. There is a need for a custom network architecture tailored to the characteristics of the control type—small but frequent packets. The issues network architects must consider include the

Defining terms

Control bus: a relatively simple mechanism for interconnecting nodes, dealing more with the physical (for example, electrical or optical) and low-level (say, error correction) attributes of the communication medium.

Control network: a means of intelligently interconnecting nodes, using both low-level and high-level concepts (say, network management and addressing).

Embedded control system: a system whose actuator outputs are controlled by a preprogrammed microprocessor or microcontroller responding to the sensor inputs.

Interoperability: the ability of products from different manufacturers to work together without modification or the use of a third-party agent (for example, gateways).

MAC: the media access control function defined by the IEEE 802 committee for local-area networks and incorporated in the data link layer of the OSI model.

Node: in a control network an autonomous processing entity usually containing a sensor, actuator, or both, plus a transceiver for communicating on a common network and a processor that runs the user application program.

Medium: the physical channel on which communication takes place.

Offered traffic: the total number of packets a network is requested to carry at any given time by all its nodes; ideally, offered traffic equals throughput (everything gets through).

OSI reference model: the International Standards Organization's (ISO's) Open Systems Interconnect architecture, a seven-layer "reference model" for computer networking. The OSI model is concerned with the interconnection of systems. The

implementation of the services at various levels, however, depends on the specific attributes of the networking application at hand.

Packet: a chunk of information passed from one node to another (or to a group of nodes) in a control network. A packet contains status or command information.

Response time: the length of time it takes for an event detected by one node in the network to cause an action to occur in another node in the network.

Sensor: a "non-intelligent" device capable of converting a physical quantity (such as temperature or light level) into electrical signals. The output of a typical sensor generally needs further processing (like analog-to-digital conversion or averaging) for use in a control system.

Throughput: the amount of information that is carried by a network at a given time. Peak and sustained throughputs refer to the capability of the network under different conditions.

Means of media access

Unlike data networks, control networks are often hit with huge bursts of data packets, dubbed "offered traffic." The media access algorithm used to handle this situation affects the response time and throughput of the system, and also the implementation cost and the cost of the node itself. Its choice is therefore a crucial step in defining a control network foundation.

Some of the media access algorithms developed for local-area networks (LANs) are obviously inappropriate for a control network's short packets (10–20 bytes), frequent occurrence (every few seconds), and high offered traffic (200–300 packets per second). In a token ring, for example, nodes must wait their turn for the token; the more nodes there are, the longer the wait; each time one is added or removed, the system must be reconfigured; and no break in the network can be tolerated.

The other popular media access techniques are carrier sense multiple access (CSMA) and master-slave.

The CSMA class of media access algorithms, as its name implies, lets a node transmit on the network only when no other transmission (carrier) is present. This by itself eliminates many possible collisions. The several CSMA transmission algorithms available behave well when the network is lightly loaded. However, each one's performance degrades at different rates as the offered

traffic is increased. The algorithms include the non-persistent, the p-persistent, and the predictive p-persistent.

With the first, upon finding a channel busy, the node backs off for a random period of time before checking for the channel status again. It repeats this process indefinitely until the channel is found to be idle, at which time the packet is transmitted immediately.

With the p-persistent type, a channel is repeatedly checked. When it is found to be idle, the packet is transmitted with probability p (a fixed value between 0 and 1). This method improves throughput as the two nodes are now less likely to transmit at the same time. The smaller p is (the larger the number of randomizing slots), the less likely is a collision, but the greater the latency overhead for each packet.

The predictive p-persistent type is like a p-persistent algorithm, except that p is dynamically adjusted to match the expected traffic on the network. For an idle or lightly loaded network, p is large (0.8 to 1). With increasing network traffic, p is reduced to compensate for the increased number of nodes that require access to the network. The method is also referred to as CSMA/CA (for collision avoidance), and is the one most suited for control networks among all the media access schemes discussed here.

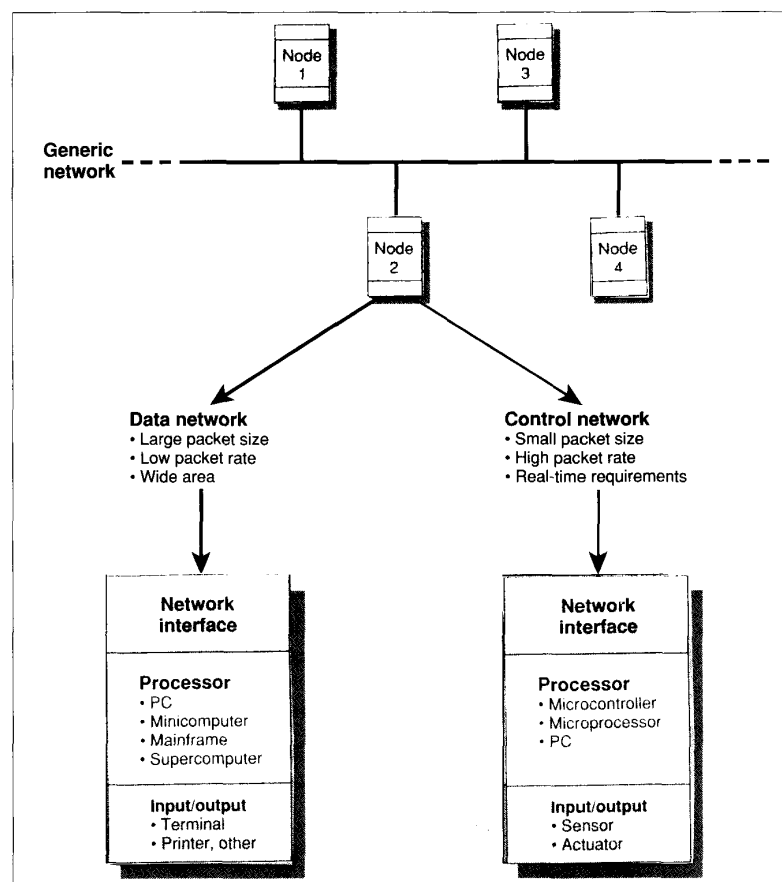
In collision resolution (CSMA/CR), all nodes that need to transmit begin transmitting their unique col-

lision resolution pattern, which is assigned by the system integrator at installation time. One bit at a time, node accesses are revoked until the node with the highest priority is left to transmit its packet. This technique is often referred to as collision resolution. A danger here is that the nodes may be locked out in the presence of persistent higher-priority node activity.

Collision detection (CSMA/CD) is popular in LAN designs such as Ethernet (IEEE 802.3). As with non-persistent CSMA, a node begins transmitting immediately after detecting an idle network. However, the nodes using this technique sense the network even after a transmission. If they detect a collision, they back off for a random time and all other nodes are alerted. This allows for faster detection of a collision than does waiting for a lack of acknowledgment from the packet's receiver.

In the master-slave method, a central node (master) polls other nodes on the network (slaves). Also referred to as polling, this technique offers a simple-to-implement protocol, and a natural transition from a centrally controlled embedded system. Some drawbacks are the presence of a single point of failure—if the master fails, the entire system comes to a halt—and bandwidth wasted by polling (especially when a slave has no new information to report). Additionally, network expansion is difficult compared to CSMA access, since the master must be updated with the new information every time.

—R.S.R.



protocol, interoperability, command status, and topology.

The network protocol must be open. Its availability to anyone anywhere on equal terms is crucial to the ability of products from different manufacturers to work together on the network. Consequently, true interoperability can be achieved only if manufacturers accept that protocol.

The cost of a node in a distributed control system is important to viability. It must be low enough to justify the transfer from an embedded control system to a distributed one and also to achieve the economy of scale possible with a large number of nodes. In the past, the cost of the silicon has been the major deterrent. Control networks are much more sensitive than data networks to cost because their application range is so much broader. A temperature sensor node in a centralized process control system may cost around US \$50. Such a node in a distributed control network might now cost \$100, due to the addition of the on-board processing and communication capabilities. The advantages gained overall and the long-term cost savings, however, usually far outweigh the initial increase in price.

[1] Although similar in many ways, data and control networks differ in the way each implements a node. The contents of a node in each case is designed to match the individual characteristics of the network at hand.

The kind of information being passed over the network is fundamental to its architecture. For a data network, this information is mainly data and occasionally network management and system control information. In a control network, the choice is less clear. Should commands be sent out from one node to specific others that dictate the behavior of the receiving nodes, or should status updates be sent out for anybody on the network to see and act on [Fig. 2]? This problem is variously handled. For example, the proposed CEBus specification is a command-based scheme where one node must command another to effect a remote action. The trouble is that with a large set of node types, this approach would require exponential expansion of the number of commands and their management overhead.

Things are simpler in a status-based network, where the functionality of each node is isolated from the rest of the network. The overall system then becomes a group of parts, each carrying out its pre-assigned tasks without direction from the others, although maybe allowing its behavior to be affected by happenings elsewhere in the network. Also, a system designer may even superimpose a command-based scheme on the status structure, in applications calling for both solutions.

TALK OF TOPOLOGY. Each of the classic network topologies for data networks [Fig. 3] has its own set of benefits and limitations. Any of these topologies will satisfy the physical wiring, performance, and cost requirements of a control network in a given application. The choice, however, is strongly swayed by two other factors: media access scheme, and the type of medium used. Together, topology, media access scheme, and medium affect virtually every aspect of the network—cost, ease of installation, reliability, and performance. The choice of topology is very often driven by the specific requirements of the overall system, such as wire installation cost or network fault tolerance. Some of the control networks under development can support two or more topologies.

Media access is crucial. Given a particular network topology, the architect must decide how the network is going to be accessed by each node. The goal is to reduce collisions among data packets and shorten the time it takes for a node to gain access to the network and begin transmitting a packet. In other words, the architect must maximize the network throughput and reduce the media access delay. Throughput is measured by the number of pieces of information that

Command-based system

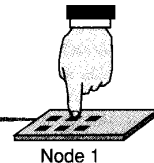
Result:
TV volume
turns up

Node 2



"Turn node 2's volume up"

Node 1



Status-based system

Result:
• TV volume turns up
• Equalizer adjusts
for new sound level
• Telephone ringer
volume increases

Node 2



Node 3

Node 4

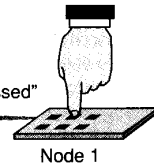
Node 5



Node 6

"TV volume-up button is pressed"

Node 1



[2] Command- and status-based control networks are not alike. The first may be viewed as an extension of a centralized system, where multiple controlling nodes operate on a common network and each must issue commands to the rest; the overall system is governed by a master list of commands. A command-based home entertainment system [top] may be a straightforward solution, given a small set of nodes.

In a status-based control system, state information is passed over the network. Each node is assumed to be an autonomous element that is smart enough to look at system state updates appearing on the

network and to decide whether to act on any of them. In a home entertainment system, this implementation would issue a message to the network in response to an input (say, when the TV volume increase button on the remote control was pressed); but this message would now contain state information conveying what had just happened [above]. Any network node may look at this piece of information and act or not act on it. Here, the television receiver and the telephone ringer choose to react by increasing their volume, while the stereo equalizer decides to adjust to the new sound level on the basis of user settings.

can be carried, using megabytes per second or packets per second for a known average packet size. Media access delay is the main factor in determining the network response time, given a network topology [see "Means of media access," opposite].

Addressing is another key factor. In a typical control network, information is originated by, and/or received from, any node. The manner in which packets of information are addressed to other nodes is important in determining the overall network efficiency and reliability. There are several approaches. In *unicast* addressing, the packet is sent to another node; in *multicast*, or group, addressing, to multiple nodes, and in *broadcast* addressing, to all the nodes on the network.

The brute force way for a node to address a message is to send it to all the other nodes on the network. This is, in fact, the way "addressing" is handled in some of the low-

end protocols, to keep communication simple and therefore lower system cost. Multicast and unicast addressing schemes are more efficient, resembling the postal system, where letters may be sent to one or more people by having a different mailing address on each envelope. Unicast and multicast addressing schemes can also permit acknowledgments and retries (like a letter sent with a return receipt requested). These features enhance the reliability and efficiency of the network, but are not practical in a broadcast-only network.

The flexibility conferred on system design by different media is well worth exploiting. Such operating conditions as data rate, distance between nodes, and reliability all affect the choice of media. In many control networks, a mix of media may be needed to satisfy the various subsection requirements [Table 1].

Routers, bridges, and repeaters offer the same sort of advantages as the connectivity tools of computer networks: support of multiple media for different network segments; easy network expansion; traffic segmentation and electrical isolation of segments to improve fault tolerance. The aim is transparent end-to-end communication without additional costs or performance degradation when integrating such tools into the network.

FAST RESPONSE. Good response time is usually demanded by the real-time control loops and proportional-integral-differential (PID) algorithms with which control applications are generally associated. For example, the delay between detecting an object on a high-speed, automated assembly line and turning on a paint-spray machine may be in the tens of milliseconds range.

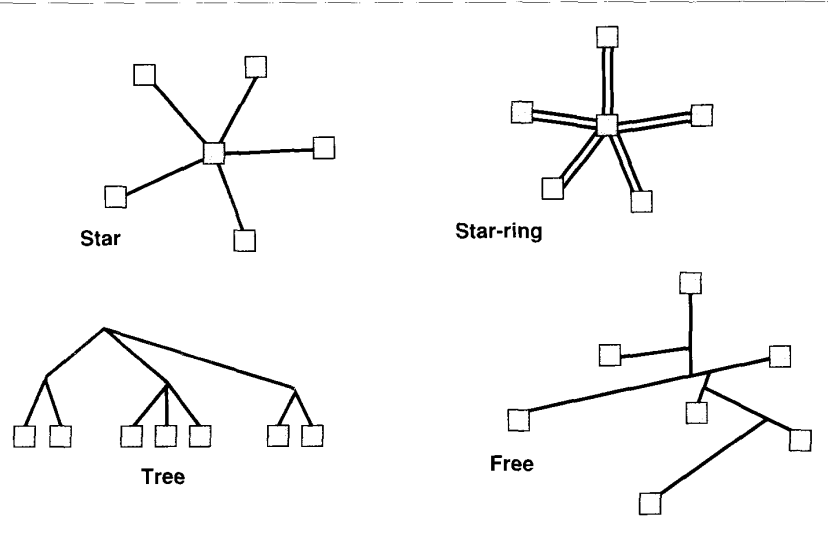
But data networks generally do not need a real-time response when sending a large data file across the network. What matters most is the media access algorithm. Next comes the overhead of implementing the layers of the OSI protocol model [Fig 4]. All seven layers need more processing power than would a simplistic system that implemented, say, only the first two layers of the OSI model. Fortunately, the benefits gained by the existence of the extra layers most often outweigh the cost factor—this has become even more apparent with advances in technology and the increasing system functionalities required by system designers.

Additionally, in an emergency or whenever speed is essential, messages must be sent with high priority. A priority mechanism is essential to bypass the normal network packet congestion.

As for security, there are two types: protection against unauthorized access and protection against system malfunction. In the first case, the password technique for security used in data networks could also be employed at the user interface nodes of the control network.

But the prospect of intruders tapping directly into the network is a different kind of threat. Their ultimate goal is not to obtain the information on the network but to abuse it in order to fool the other nodes into obeying the intruder's commands (for example, to disable the alarm system).

Authentication is one way to protect a



[3] The topologies used for data (computer) networks could as well be used by control networks, although some are better suited to the latter type's inherent characteristics. For example, a free topology may reduce wiring cost when a home is being retrofitted with a control network. (Not shown are a ring and a bus.)

node from unauthorized nodes. In essence, the system designer or integrator assigns each node a unique key. Upon receipt of any information, the node uses its key to transform a particular packet. The results are then compared, and if they are equal, the receiving node proceeds with processing the incoming data. This scheme ensures that the recording and later playback of network messages by an intruder will not allow access to the network. The price—the extra processing overhead—is seldom an issue, as only a few nodes in a system would require such protection.

To secure a control network against malfunctions, the system designer has many options. The inherently distributed nature of control networks tends to eliminate single points of failure. Also, redundant nodes could be used at vital points, or else a monitoring node might continually query the other nodes for a healthy response.

In the area of network management, the control network is typically demanding because of its frequent field reconfiguration and upgrades. Network management tools are used to define node addresses and the relationships between the nodes; to add new nodes and replace defective ones; and to

test any node and any function needed to properly start up and maintain the network. Network management comes as a surprise to most people moving from centralized embedded systems to control network design. The earlier system never had to replace nodes or address them. Relationships between its "nodes" (sensors and actuators) were determined by some of the central control program's algorithms, usually programmed into the system, as in programmable logic controllers. Changes to the system, like adding new sensors and actuators, therefore required costly changes to the system wiring and the involvement of a knowledgeable system installer/programmer. But the "necessary evil" of network management is really an enabling tool to harness the new capabilities of control networks, including the autonomy of nodes, the flexibility of configuration, and the ease of maintenance.

Network management requirements vary with whether the network is pre-installed, self-installing, or installed in the field. The pre-installed type is installed during manufacture, usually being embedded in products that will have a fairly stable configuration in the field. For example, the nodes in an intelligent automobile (such as

1. Distributed control media

Characteristics	Twisted pair	Radio	Power line	Coaxial	Infrared	Fiber optics
Typical range, meters	1-1000	50-10 000	10-5000	10-10 000	0.5-30	10-10 000
Typical data rate, kb/s	0.3-2000	1.2-9.6	0.06-10 000	300-10 000	0.05-20	1-100 000
Relative node cost, US \$	\$10-\$30	\$50-\$100	\$50-\$150	\$30-\$50	\$20-\$75	\$75-\$200
Typical installation cost	Low	—	None-low	Medium	—	Medium-high

OSI layers		Typical task assigned	Control networking requirements
7	Application	• User semantics • File transfer	• Data objects • Standardized networking structures
6	Presentation	• Data compression • User data conversion	• Networking structures • Data interpretation
5	Session	• Synchronization • Dialog structure	• Authentication • Network management
4	Transport	• Reliable data transfer • End-to-end communication	• End-to-end acknowledgment • Duplicate detection, automatic retries
3	Network	• Routing, logical addressing • MAC-independent interface	• Addressing, unicast, multicast, broadcast • Routers
2	Data link	• Media access scheme • Error correction, and framing	• MAC, collision avoidance/detection • Framing, data encoding
1	Physical	• Physical interface definition • Transceiver interface	• CRC, error checking • Priority • Media transceivers

MAC = media access control CRC = cyclic redundancy

[4] Of the many standards and models that have been proposed for data networking, the International Standards Organization's Open Systems Interconnect (ISO's OSI) is perhaps the most robust and generic. Certainly its international popularity is considerable. OSI is a seven-layer reference model for computer networking; but it is applicable to any kind of network in which the processing elements communicate over a common path.

The above diagram shows an OSI implementation tailored to the needs of control networking. In essence, it represents a road map for designing a robust control networking platform of a kind applicable to many industries and applications. Sometimes a subset of layers is implemented to produce a simple solution for "quick and dirty" control networking. This subset is referred to as a control bus, and usually represents a specification on one or two layers (usually physical and data link) of the OSI model.

What is gained by a control bus approach is a less costly implementation because fewer layers are provided. For most real applications, though, a more nearly complete subset of the OSI layers is needed. In theory, the extra layers needed could be implemented on top of a control bus definition. In practice, the result suffers from difficulty of implementation and lack of interoperability.

security, lighting, navigation, and temperature control nodes) would be pre-installed in the factory. Any later changes could easily be made by a qualified mechanic using a factory-supplied tool.

Self-installing networks allow some user involvement. In a home lighting system, for example, several nodes (switches, dimmers, lamps, motion detectors) would be installed throughout the house, each with a simple user interface (maybe a dial), which would be set to suit the location (a bedroom, say) and its logical connection (maybe the ceiling light). A self-installation option allows for easy reconfiguration of the system by the end user without the need for any complex tools.

Field installation requires a special tool, used in the field, to assign node addresses and logical node interconnections. The installation tool can be as simple as a handheld unit with a simple user interface, or as sophisticated as a PC running a graphical user-interface. Field installation permits total control over the network but requires a level of proficiency and active involvement on the user's part that may not always be available or cost effective.

Since most installation scenarios combine two or all of the above options, the control networking protocol must be able to support all three options. The protocol must also support

functions for testing nodes over the network.

Finally, monitoring and control is a natural extension of the control networking concept. In a factory assembly line, for instance, a monitoring and control station in the main office could give the factory supervisor status updates on every assembly station. In turn, the supervisor could control network functions, such as process rates and emergency shutdown, through this station. The monitoring and/or controlling node may be thought of as just another network node. This "super node" could be added or removed at any time, without disrupting the rest of the network. In a centralized control system, in contrast, the monitoring and control functions must be built in from the start.

WHAT'S OUT THERE. Numerous groups have considered these issues and proposed control network "standards" for specific industries. Many

of these standards are still evolving, and some are being combined with others to "migrate" to another set of proposed standards. They fall into three categories: a *proposed* standard, in which the full specification has not been completed, and therefore no truly interoperable products yet exist; an *instance* of a draft standard, which is a complete specification with enough undefined or unclear areas for each adoption of it to have yielded a different variation, and therefore interoperability is virtually impossible; and an *implemented* standard, which is complete, fully specified, and implemented in the form of a chip or chip set (including firmware and software), and can therefore be used to produce interoperable systems.

Choosing a protocol is most often a trade-off between the advantages in hand and the wait for a useful and stable specification. Some users therefore combine various offerings—to illustrate, three protocols, BACnet, CAB, and LonWorks, were recently adopted by the Intelligent Buildings Institute, Washington, D.C., a trade association of manufacturers, designers, construction companies, and others involved with creating smart facilities.

The table lists some existing protocols from around the world, each at a different stage in its evolution [next page]. They are representative of protocols in each of the main markets: building automation, auto-

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2. Leading architectures for distributed control*

	BACnet	CAN	CEBus	IEEE-488	ISP	LonWorks	WorldFIP
Characteristics							
Application(s) targeted	Building automation	Automotive	Consumer	Instrumentation	Process control	All	Process control
OSI layers	1,2,3, 7	1,2	1,2,3, 7	1,2, 7	1,2, 7	1,2,3,4,5,6,7	1,2, 7
System control (command- or status-based)	Both	Command	Command	Command	Status	Both	Both
System type	Network	Bus	Net	Bus	Bus	Net	Bus
Media access	CSMA/CD, token bus, master-slave, dial-up	CSMA/CR	CSMA/CD	Bus	Master-slave, token passing	CSMA/CA	Master-slave, token passing
Error correction	CRC	CRC	CRC (only power line)	—	CRC	CRC	CRC
Media supported besides twisted pair	Coaxial cable, optical fiber	Fiber	RF, power line, coax	(Twisted pair only)	(Twisted pair only)	RF, power line, IR, fiber, coax	Fiber
Addressing schemes (uni-, multi-, broadcast)	All	Broadcast	All	Unicast, broadcast	All	All	Broadcast
Maximum data rate, Mb/s	10	1	0.01	8	2.5	1.25	2.5
Intrinsic safety?	No	No	No	No	Yes	Yes	Yes
Power from network?	No	No	Yes	No	Yes	Yes	Yes
Max. no. of nodes	2 ⁴⁸	40	2 ¹⁶	961	8128	2 ⁴⁸	256
Security	Authentication, encryption	—	—	—	—	Authentication	—
Priority	Yes	Yes	Yes	No	Yes	Yes	Yes
Support							
Network management?	No	No	No	No	Yes (a)	Yes	Yes
Connectivity (repeaters, bridges, routers)	All	None	None	Repeaters	Bridges	All	Repeaters
Current availability							
Chip or chip set?	No	Yes	No	Yes	No	Yes	Yes
Tools:							
Node development?	No	Yes	Yes	No	Some	Yes	Some
Network development?	No	No	No	No	No	Yes	No
Protocol analysis?	No	No	No	No	No	Yes	Yes
Network management?	No	No	No	No	No	Yes	No
Connectivity?	No	No	No	Yes	No	Yes	No
Training?	No	Yes	Yes	No	Some	Yes	Yes
Support?	No	No	No	No	Some	Yes	Yes

* For full names of architectures, see text below and opposite.

(a) Defined in application layer.

CSMA = carrier-sense, multiple-access; CA = collision avoidance; CD = collision detection; CR = collision/resolution; CRC = cyclic redundancy check; OSI = Open Systems Interconnection.

motive, home automation, instrumentation, factory automation, and process control.

BACnet stands for building automation and control network. It is being drafted by the American Society of Heating, Refrigeration, and Air-Conditioning Engineers (ASHRAE) to provide a standard communication and environmental control

network for commercial and government buildings and campus environments. So far, only HVAC (heating, ventilation, and air-conditioning controls) have been designed in.

The BACnet protocol, still under development, is defined to operate on any of the acceptable media for Ethernet and Arcnet, as well as the RS-485 twisted pair. A point-

to-point data link layer is also specified for modem dial-up services.

The standard is expected to be easy to extend to new HVAC control technology and other building services, as well as to work with other communication protocols such as Novell's Netware and Echelon's LonWorks.

CAN is short for controller area network.

Designed jointly by Bosch, Philips Components, and Intel for the automotive industry, the protocol specification focuses primarily on the data link layer of the OSI model, thus making it more of a bus than a network definition. Application message structures must be implemented by the user, as CAN simply provides the frame format for rudimentary routing.

Several CAN chips are available from Intel and Motorola, Philips and Siemens. Some provide the basic CAN core module, while others include a processor as well in the same package. In every case, the user implements the remaining layers of the protocol stack to match the application at hand.

CAN aims at being a simple and relatively high-speed bus with good immunity to the errors endemic in the harsh automotive environment. It has gained some acceptance in the automotive and truck industry in Europe, in addition to limited use in industrial automation. The tepid welcome is due to the high cost of a complete node implementation.

CEBus, or consumer electronics bus, is proposed by the Electronic Industries Association (EIA) and intended for consumer applications. It centers on the physical, data link, and application layers of the ISO OSI model.

CEBus is a draft interim standard, IS-60, in development for nine years and an example of a command-based specification. No available ICs readily implement the CEBus standard. The user must implement the application program on a microprocessor and make sure that the product is compatible with implementations from other manufacturers.

IEEE 488, also known as the general-purpose interface bus (GPIB), is a bus standard popular with manufacturers of sophisticated laboratory instrumentation and readily implemented on several chip sets. It does not lend itself well to types of control other than instrumentation, however. Also, the number of devices on the bus is limited to 15 (although repeaters, could increase this number to 961).

THREE MORE TO GO. Interoperable Systems Project (ISP) was formed by Fisher-Rosemount, Siemens, Yokogawa Johnson, and Foxboro for process control. Manufacturers cannot agree on a field bus standard, and the project's goal is to propose one that would ultimately unite the benefits of several specifications.

ISP combines the physical layer definition of the Instrument Society of America's SP50 specification with enhanced data link and application layer definitions of the Profibus (Process Field Bus). So far, no chips support the ISP specification, but some should become available within the next few years. These physical and data layer chips would then need to be interfaced to a host processor containing the user application code.

LonWorks, from Echelon Corp., Palo Alto, Calif., is applicable to a wide range of industries. It provides services at all seven layers of the OSI model, along with application

development tools and a user application processor, in a single device called the Neuron chip. The IC is manufactured and sold worldwide by both Motorola and Toshiba. Implementation of a complete node requires only the addition of sensors and actuators.

The LonWorks specification supports most media. Currently supported are twisted pair, power line, radio frequency, infrared, coaxial cable, and optical fiber. The LonWorks products include a set of solutions for intelligent distributed control, such as network management tools, development tools, OEM modules, and connectivity tools (such as routers, and network interfaces).

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By designing a multipurpose platform for rapid development of control networks, LonWorks may not be suitable for use in extremely high-speed applications requiring sub-millisecond response times, like automotive engine control. This is an issue with all the control networking standards discussed once the protocol required for a realistic network has been added. Over 800 companies worldwide have adopted this standard.

Finally, the world factory information protocol (WorldFIP) is based on the FIP, originally from Télémécanique of France, and the Instruments Society of America (ISA) field-bus proposal, SP-50, and is intended for the discrete manufacturing industry. Like ISP, WorldFIP is meant to provide a universal field-bus solution for the process control and factory automation industry.

At the user level, WorldFIP employs a subset of the Manufacturing Messaging Service of the Manufacturing Automation Protocol (MAP), originated by General Motors Corp. for its factories, and incorporates a communication model called the producer/distributor/consumer model, which translates into a broadcast-only messaging service.

Several custom application-specific ICs provide different parts of the protocol specification. They form the physical communication interface to the network and require an external processor for the user application.

TO PROBE FURTHER. No off-the-shelf texts deal specifically with control networking. But, Andrew Tanenbaum's *Computer Networks* (Prentice Hall, Englewood Cliffs, N.J.) and John Spragins' *Telecommunications, Protocol and Design* (Addison Wesley, Reading, Mass.) provide excellent background information on general networking concepts.

For more on developments in data net-

working, see "ATM knits voice, data on any net" by James Lane, *IEEE Spectrum*, February 1994, pp. 42-45. *IEEE Network: The Magazine of Computer Communications*, published by the IEEE Communications Society, covers this topic regularly.

As for the control networking proposals discussed, the available literature ranges from minimal to very extensive.

- For BACnet, copies of the latest Advanced Working Draft 3 can be obtained from the American Society of Heating, Refrigeration, and Air-Conditioning Engineers (ASHRAE). (Call 404-636-8400 for SPC-135P-031)

- For CAN, bus controller chips include Intel Corp.'s 82527 (call 408-765-8080) and Motorola Inc.'s MC68HC05X4 (call 512-873-2000). The CAN specification, Version 2, is published by Robert Bosch GmbH, Stuttgart, 1991.

- For CEBus, EIA IS-60 CEBus Specification, Vols. 1-7, October 1992, may be obtained by calling the Electronic Industries Association (EIA) in Washington, D.C. (202-457-4900).

- For IEEE 488, the two common GPIB chips are NEC Corp.'s μ PD7210 and Texas Instruments Inc.'s (TMS9914A). The official 1987 specification for IEEE 488 may be obtained from the IEEE. Call 1-800-678-IEEE and ask for SH 11346-PIN.

- For ISP, "Revision 3 of the ISP Field Bus Specification" can be obtained from the ISP Foundation, Austin, Texas (512-794-8890).

- For LonWorks, ask for "LonTalk Protocol," April 1993, in the *LonWorks Engineering Bulletin*, from Echelon Corp. For details on learning LonWorks technology, contact the Palo Alto, Calif., company at 1-800-258-4566 or 415-855-7400. For information on the Neuron Chip, contact a local Motorola or Toshiba distributor.

- For WorldFIP, contact WorldFIP N.A. Inc. in Research Triangle Park, N.C., at 919-941-5121.

Additionally, several comparative analyses exist, including study reports such as: the Gas Research Institute's "Home Automation Systems in North America: An Analysis of the Three Main Contenders," September 1993, obtainable from the Commerce Department's National Technical Information Services at 703-487-4650 (ask for PB94-118031); "Echelon Neuron vs. Microcontrollers: Side by Side Comparison," October 1991, Microsym Computers, 131 Finchdene Square, Scarborough, Ont., Canada M1X1A6 (416-293-8263); and "Embedded Communication Protocol Options," B. Upender and P. Koopman, *Proceedings of the 2nd Annual Embedded Systems Conference*, Vol. 2, April 19-21, 1994, pp. 221-32. ♦

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